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Looking back on this project, I feel really confident about my programming skills with Arduinos and our team's ability to build a functioning prototype from scratch. Our project was designed to serve as a working model of an automated humidity awning, combining electrical engineering with practical structural design. The humidity sensor tracks resistance changes in the air to detect shifts in humidity, and when a substantial change occurs, the Arduino triggers a green LED to turn on high as a warning light that weather conditions are changing. This gives the property owner time to move items indoors or clear the area underneath the awning before rain arrives. Meanwhile, the moisture sensor detects actual water by registering a noticeable change in voltage, which causes the Arduino to trigger a servo motor to pull out a piece of paper, deploying it outward to act as a protective rain cover until the sensor dries off and the servo retracts it.

Beyond the technical hurdles of wiring the circuit and writing the code, one of the most valuable lessons I learned was the importance of workflow order, specifically, to work on the insides before working on the outsides. It is much easier to custom-build a case to fit your circuit than it is to try and squeeze finished electronics into a tight, pre-made box, so we were able to precisely craft a cardboard and popsicle stick house around our electronics rather than forcing components into a cramped space. Finally, this project helped me grow significantly as a collaborator, teaching me that keeping constant, clear communication with my teammates was essential for troubleshooting code, coordinating the wiring with the physical structure, and ensuring everyone stayed on the same page.